efficiently; only the required data must be retrieved (usually from a database back-end).

- **Database:** Continuing from the point above, all the major frameworks today support a large number of available database products, from a variety of vendors like MySQL, Oracle, Microsoft SQL Server, etc. Database functionality for the Web application is exposed by the framework in its API, enabling functions like adding or deleting records, fetching data, and the like.
- **User management:** This is a very important feature in a framework. A Web application must support user logins, roles and all access-related features, which are powered by the framework's security API.

Behold! Architecture

A cornerstone in the concept of frameworks is the architecture. Taking forward the example of the cube, as in any structure, architecture is an important aspect. From the point of view of the Web application framework, architecture means the way the framework is built and developed, a popular example being the MVC (Model View Controller) architecture. There are books on the subject, so explaining it completely is beyond the scope of this article. In short, the speciality of this architecture is that it splits the framework into three blocks or parts: the Database Model, the User Interface (View) and the Business Logic (or controller—the brains of the system, the glue that holds the other two blocks together).

Popular frameworks

I will list three popular Web application frameworks called Content Management Systems (CMSs). A CMS is a high-level framework that is used for rapid application development (RAD). If you need to quickly deploy a website/Web application, a CMS is the way to go. It has all the pieces in place and is ready for customisation. It has an inbuilt control panel, administration sections, configuration panels and guidelines to skin and plug in more functionality. As the name suggests, it manages content or data as an inherent feature.

Joomla!

The Joomla! Project has its roots in Mambo, the product of a company called Miro Construct Private Limited. The Mambo CMS, initially a closed-source proprietary system, is now licensed under the GPL.

Code platform: PHP
Database back-end: MySQL

Drupal

The brainchild of Dries Buytaert, Drupal (in Dutch, druppel means 'water drop'), is another popular CMS. It has a large and ever-growing community, along with user groups that have been formed around the world.

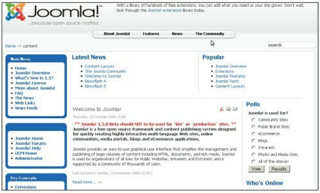

Figure 1: Joomla

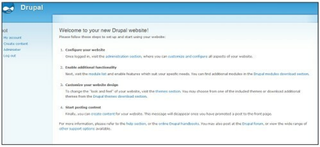

Figure 2: Drupal

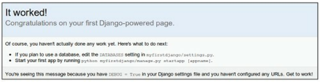

Figure 3: Django

Code platform: PHP
Database back-end: MySQL/PostgreSQL/SQLite

Django

Developed by Lawrence Journal-World and released under the BSD License, this CMS is named after the jazz musician Django Reinhardt. It's built purely using the MVC architecture, and emphasises the DRY (Don't Repeat Yourself) software engineering principle, which, to put it simply, reduces the redundancy of data/information/code in the system.

Code platform: Python
Database back-end: MySQL/PostgreSQL/SQLite/Oracle

The depth is infinite in the world of frameworks, as it is in knowledge everywhere. It's time to pause and ponder over all this information that has emerged. Now that 'Framework' is no longer an alien term to you, go ahead and explore! May the frameworks be with you. END

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